Shubhankar Kumar

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Summary

Accomplished with a Master's degree in Mathematics and Scientific Computing, equipped with strong skills in artificial intelligence and machine learning. I am passionate about applying my expertise in computer vision, deep learning, and data analysis to contribute to impactful projects.

Experience

•Code Clause *Fab 2023 to March 2023*
Artificial Intelligence Intern Live Url |

– Working on cutting-edge technology in object detection, object tracking, OCR in natural scenes, image classification, video analysis, and many more exciting techniques.

•IBM *Dec 2022 to fab 2023*
Artificial Intelligence Intern Live Url | Github

– Experience with big data, deep learning frameworks and cloud computing platforms, like Spark, CUDA, TensorFlow, Keras, OpenCV, Azure, AWS, GCP.

•The Sparks Foundation (A Non-Profit Education Management Industry based on Singapore) *July 2022 - Aug 2022*
Data Science Intern Live Url

– Experience in advanced data analytics and visualization tools including Microsoft Office Suite, Power BI/Tableau, Flask, TensorFlow,SQL,Machine learning, and gain knowledge of data management/warehousing etc.

•SORADIS GLOBTECH PRIVATE LIMITED *Mar 2022 - May 2022*
Junior Web Developer Live Url

– Worked on Following Technologies-HTML,CSS , JavaScript, MySQL, Git.– Web Scrapping using Python Libraries-Beautiful Soup , Selenium , CSS Selector , Scrapped educational website and gather Data in xls format..

Projects

•Human Emotion Detection : Comprehensive Deep Learning Approach *Jan. 2024*
Github Url |

Implemented a **comprehensive deep learning approach** with models like **LeNet, ResNet34, EfficientNet, MobileNetV2, ViT, and HuggingFace ViT**, achieving accuracies up to **89.29%**. Conducted benchmarking on **TensorFlow** and **ONNX**, showcasing GPU-optimized models with inference times as low as **0.025s**. **TensorFlow GPU** model demonstrated **0.15s** inference, **0.8s** CPU time, and **1000MB** size. **ONNX GPU** model showed **0.025s** inference, outperforming TensorFlow, with a reduced model size of **328MB**.

•PharmaOptiMix: Pharmaceutical Sales Optimization Project *Dec. 2023*
Github Url |

Conducted pharmaceutical sales forecasting using diverse models, including **ARIMA**, **Auto ARIMA**, **Prophet**, and various **LSTM** architectures (**Vanilla**, **Stacked**, **Bi-Directional**), achieving a notable **17.25% Mean Absolute Percentage Error (MAPE)**. Demonstrated expertise in both traditional and modern time-series analysis methods, surpassing **Naïve methods** for improved accuracy. Proven track record of optimizing hyper-parameters and contributing to data-driven insights for strategic decision-making in the pharmaceutical industry. Instrumental in informing resource planning and marketing strategies with a forward-looking approach.

•PubMed 200k RCT Sequential Sentence Classification With SkimLit *Nov. 2023*

Github Url |

Our exploration of deep learning models for sequential sentence classification in medical abstracts, leveraging the PubMed 200k RCT dataset, yielded **promising outcomes**. The **BERT-based model** stood out with an impressive accuracy of **88%**, surpassing other architectures. Notable performances were observed across various models, including the **NaiveBiase Model (72% Accuracy), Conv1D Model (78% Accuracy), Pretrained Token Embedding (Universal Sentence Embedding - 75% Accuracy), Conv1D Model using Character Level Embedding (73% Accuracy), Model with Both Token and Character Level Embedding (76% Accuracy), and Model with Token, Character, and Position Level Embedding (81% Accuracy)**. These findings emphasize the efficacy of **advanced embeddings** in capturing nuanced patterns in medical abstracts. Our success paves the way for future research, including generalization assessments and the exploration of **ensemble methods** for further improvements. We extend our gratitude to the authors of the "PubMed 200k RCT" paper, whose dataset and insights significantly contribute to advancing **NLP in the medical domain**.

•AI-driven Conversational Q&A System for ShubH Tees *Oct. 2023*

Github Url |

Designed and implemented an innovative conversational interface leveraging **Google Palm LLM, Hugging Face embeddings**, and **Streamlit** for seamless interaction with ShubH Tees' **MySQL** database. Integrated advanced technologies such as **Langchain framework, Chromadb** vector store, and **Few Shot Learning** for precise natural language query handling. Demonstrated expertise in **NLU/NLG**, resulting in a feature-rich system with **Streamlit UI**, **Langchain** integration, and efficient **Chromadb** vector storage. Successfully delivered a cutting-edge Q&A solution, showcasing diverse technical skills and innovative problem-solving..

•Shubh Chatbot For Food Delivery For Restaurant Service *April. 2023*

App Url |

Developed a versatile restaurant services chatbot using Dialogflow, with a focus on menu information, reservation bookings, and customer support. **Menu Info:** Dishes, prices, and dietary details.**Reservations:** User-friendly booking system. **Order Tracking:** Real-time updates on food orders.**Special Offers:** Notifies about discounts. Developed a robust restaurant chatbot using Dialogflow, integrating backend systems for reservations and order processing. Recognized for enhancing user experience through intuitive features and real-time services..

Education

Degree/Certificate	Institute/Board	Year
PGP in DS, BI and AI	Aegis School of Data Science	2023-2024
Master's in Mathematics and Sc.Comp	National Institute of Technology , Allahabad	2021-2023

Skills

- **Programming**: Python, C/C++
- **Artificial Intelligence**: Q-Learning, Deep Q-learning, A3C
- **ML Algorithm**: Supervised, Unsupervised, Reinforcement
- **Deep Learning Algorithms**: ANN, CNN, RNN , Computer Vision
- **NLP**: Text Normalisation (Stemming and Lemmatization), Sentiment Analysis, Word Embedding , LSTM RNN ,BERT ,Transformer
- **Web Framework and Cloud Deployment**: Flask, Fastapi, Heroku , AWS* ,GCP*,Azure*, Spark ,Hadoop*
- **Tools/Frameworks**: Pandas , Keras, Numpy , Matplotlib ,Tensorflow, Pytorch, SciPy, OpenCV ,
- **Operating Systems**: Windows, Linux*

** Elementary proficiency*

Key courses taken

- **Computer Science**: Computer Programming (with Lab) , Data Structures and Algorithms (with Lab) , Database Management System , Machine Learning , Deep Learning , Artificial intelligence , Computer Vision.

Achievements

- **IIT JAM 2021**, IIT JAM Entrance Qualifier conducted by IISc Bengaluru, **AIR-802** *2021*
- **GATE 2022** , Secured **AIR 1007** among 0.15 million candidates appearing for the test *2022*
- **13th Edition AGBA Winner 2023**, Aegis Graham Bell Awards is one of the most credible and largest innovation awards supported by the Ministry of Electronics and Information Technology, Government of India - NITI Aayog; Skill India - National Informatics Center Services Incorporated. *2023*